BTS Airline On-Time Performance 2024: A Six Sigma DMAIC Analysis

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Dataset Overview

Dataset: BTS Airline On-Time Performance 2024
Scope: 6,965,266 flights, 15 carriers, 348 airports

1 Introduction

Flight delays represent one of the most persistent operational failures in U.S. commercial aviation. They erode passenger satisfaction, generate significant costs for airlines through crew overtime, fuel burn, and gate conflicts, and trigger cascading schedule disruptions across entire fleets. This project applies the Six Sigma DMAIC framework to the full-year 2024 BTS Airline On-Time Performance dataset - 7,079,081 raw domestic flight records spanning 15 carriers, 348 airports, and 6,792 unique routes, to quantify the delay problem, isolate its root causes through statistical analysis, and develop a data-driven improvement and monitoring strategy.

2 Define Phase

Scope & SIPOC: The study maps the end-to-end departure process. Suppliers include airlines, ground crews, ATC, and fuel vendors. Inputs are flight schedules, crew rosters, airspace slots, and gate assignments. The core Process covers five stages: check-in, boarding, pushback, taxi-out, and takeoff, producing Outputs (departure records, delay values) consumed by Customers: passengers, airline ops centres, and regulators.

A flight is a defect if its departure delay exceeds 15 minutes. This threshold underpins all sigma calculations and capability assessments throughout the study.

3 Measure Phase

3.1 Data Preparation

After removing 113,815 records with missing values and dropping the cancellation code column, the working dataset comprised 6,965,266 flights.

3.2 Baseline Performance

Table 1: Baseline Performance Metrics

Metric	Value
Mean departure delay	12.59 minutes
Median departure delay	−2.0 minutes
Flights delayed > 15 min	1,388,886 (19.94%)
Overall DPMO	199,402
Industry Sigma Level	2.34σ (Poor)

The distribution is strongly right-skewed: the median flight departs slightly early, but the mean is dragged to 12.59 minutes by a heavy tail of severe delays. The 99th percentile stands at 215 minutes, with a maximum of 3,777 minutes, confirming that average-based reporting substantially understates the true delay burden.

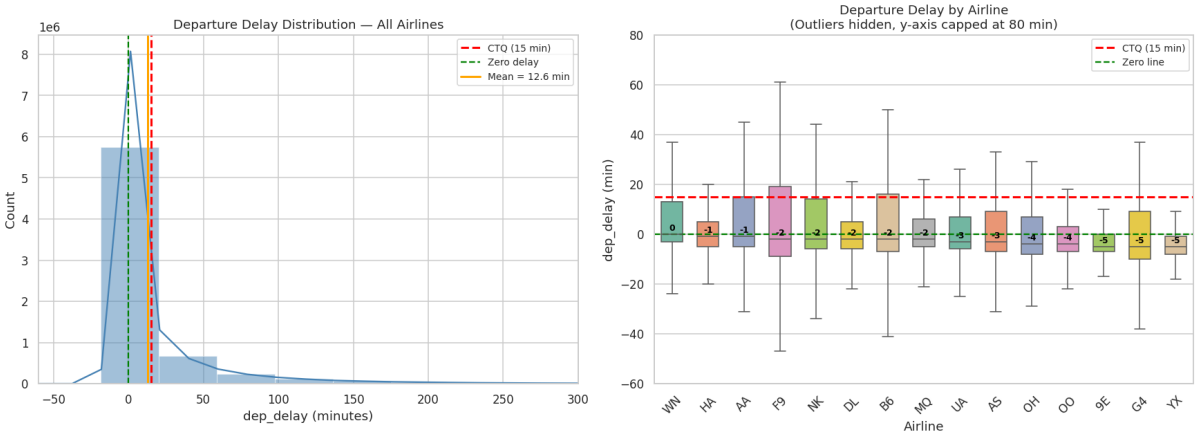


Figure 1: Departure Delay Distribution

3.3 Per-Airline Sigma Performance

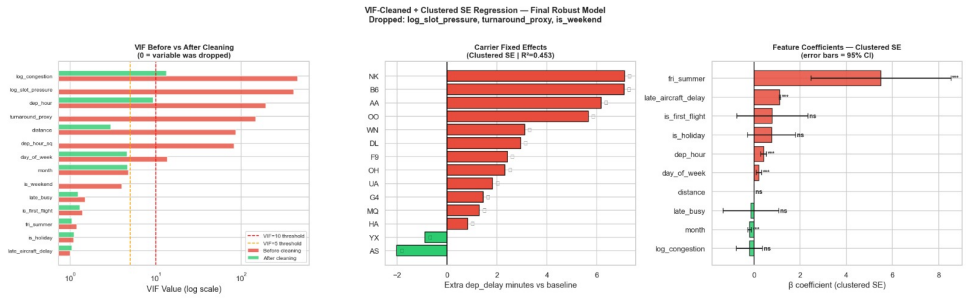


Figure 2: Per-Airline Sigma Levels

Sigma levels ranged from 2.71σ (YX, Contour Air - best) to 2.11σ (F9, Frontier - worst). Critically, every single carrier fell below 3σ, and process capability (C_{pk}) was below 0.1 for all airlines, confirming this is a systemic, industry-wide failure rather than an

isolated carrier problem. The gap between best and worst is only 0.6σ, further supporting a structural interpretation.

3.4 Temporal Patterns

Fridays average 14.6 minutes of delay (worst day); Wednesdays average 10.3 minutes (best). Seasonally, July peaks at 22.2 minutes average and October is the best at 5.4 minutes, a swing of 16.8 minutes representing the single largest controllable temporal pattern in the dataset.

4 Analyse Phase

4.1 Pareto Analysis

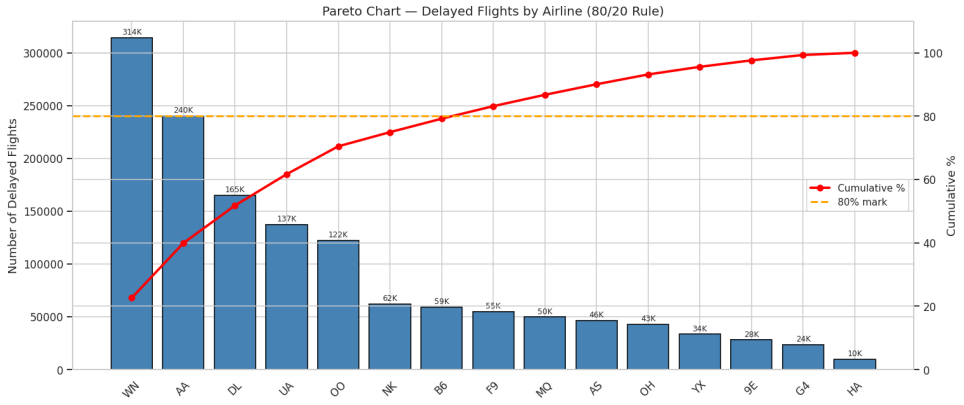


Figure 3: Pareto Analysis of Delay Contributors

The Pareto analysis confirmed the 80/20 rule with striking clarity. Just 6 of 15 airlines (WN, AA, DL, UA, OO, and NK), account for approximately 75% of all delayed departures. Southwest (WN) alone contributes 22.6% of all delays in absolute volume (314,264 delayed flights), and the top two carriers (WN + AA) together account for nearly 40% of industry delays. This concentration directs improvement resources to a well-defined subset of carriers.

4.2 Root Cause Analysis — Delay Cause Attribution

74.9% of all delay minutes are controllable, strongly supporting the case for a structured improvement programme. The cascade effect is the most critical finding: the correlation between a flight's departure delay and the late-aircraft delay it imposes on the next rotation is $r = 0.639$, while departure delay to arrival delay is $r = 0.971$ — nearly deterministic. Southwest (WN) has the highest cascade correlation at $r = 0.774$, followed by Frontier (F9) at $r = 0.749$. This means fixing the first delay of the day stops a chain reaction through the rest of the schedule.

We have also traced some of the root causes of airline delays, and attributed them in an Ishikawa diagram (see Fig. 4).

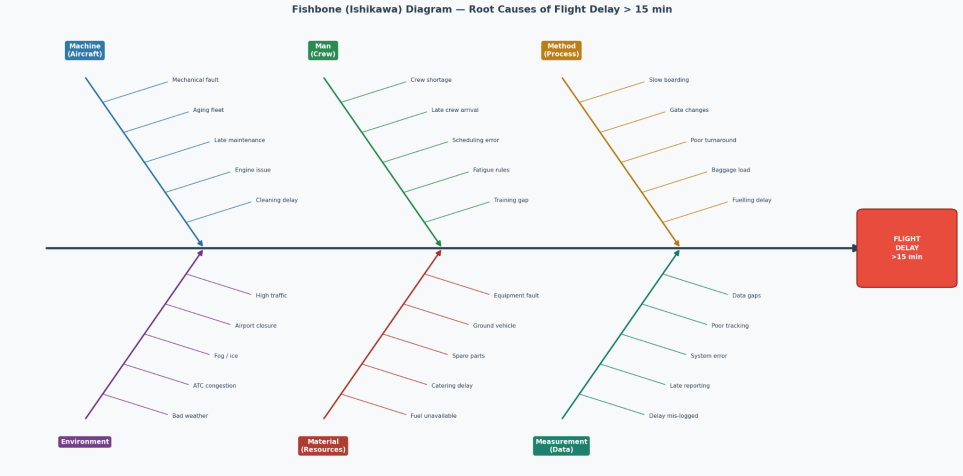


Figure 4: Ishikawa Diagram - Root Causes of Delays

Table 2: Delay Cause Attribution

Cause	**Nature**	**Share**
Late Aircraft (cascading)	Controllable	40.40%
Carrier Delay	Controllable	34.50%
NAS / ATC Congestion	Partially controllable	18.90%
Weather	Uncontrollable	6.00%
Security	Negligible	0.20%

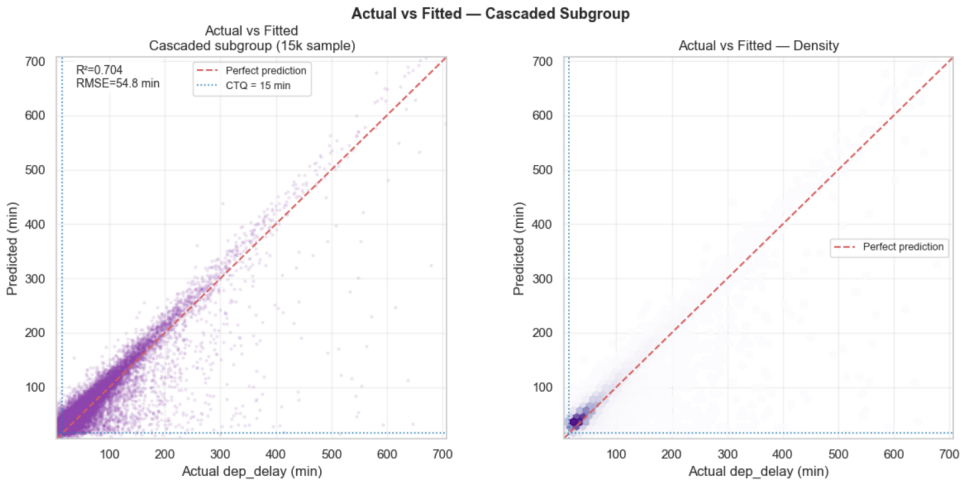


Figure 5: Cascade Correlation Analysis

4.3 Statistical Hypothesis Testing

Three formal tests were conducted to confirm that the patterns observed are not attributable to chance:

H1 — ANOVA across airlines: $F = 2,854.70$, $p < 0.0001$. The null hypothesis is rejected; departure delay means differ significantly across carriers. Tukey HSD post-hoc testing identified 49 of 105 pairwise comparisons as statistically significant, meaning some airlines are operationally indistinguishable in delay performance and can be benchmarked against each other.

H2 — Kruskal-Wallis across days of week: $H = 28,168$, $p < 0.0001$. The Friday peak (14.6 min) versus Wednesday trough (10.3 min) is a genuine operational pattern, not sampling noise.

H3 — Kruskal-Wallis across months: $H = 147,540$, $p < 0.0001$ – the strongest signal of the three. The July–October swing of 16.8 minutes is the largest statistically confirmed pattern in the dataset.

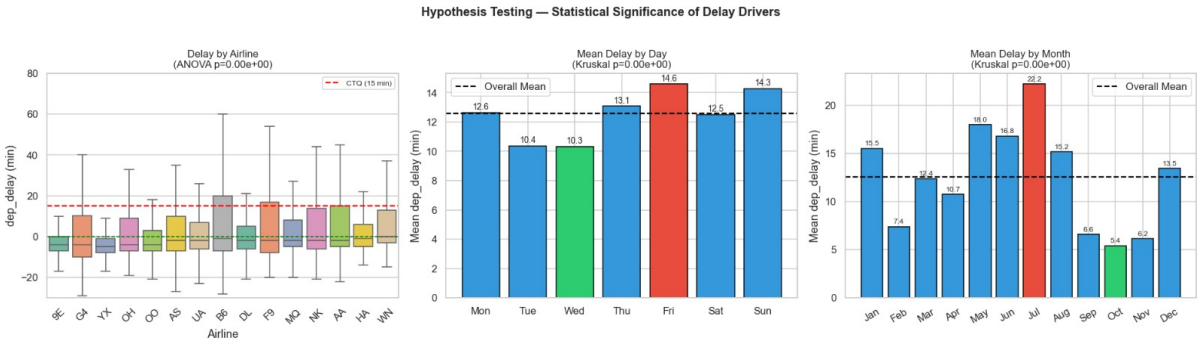


Figure 6: Statistical Hypothesis Testing Results

4.4 Multivariate OLS Regression

To evaluate the efficacy of this approach, we initially applied a Multivariate OLS Regression model to the Cascaded Subgroup—flights characterised by a pre-existing late aircraft delay ($LAD > 0$). We turned a noisy environment into a very predictive one by separating this high-risk group from the full dataset of almost 7 million flights. The most important thing is that the model's explanatory power has improved: we achieved an R^2 of 0.7044, indicating that it now explains more than 70% of the variation in departure delays for these flights. The model trained on the full dataset has R^2 of 0.4874, thus our choice in focusing on the cascaded subgroup is a significant improvement over this naive approach.

We used a Z-test to determine whether the late_aircraft_delay coefficient (β) differs when the cascade is present, which is how we mathematically tested this strategy.

We clearly rejected the null hypothesis because the Z-statistic was -13.367 and the p-value was 0.0000. This strong statistical evidence shows that the relationship between inherited delays and departure times differs in the cascaded group. We moved away from a generalised average toward a specialised framework that better captures the complex dynamics of delay propagation, showing that "late planes follow different rules." This ensures that our coefficients are not merely correlations but statistically significant drivers of the outcome.

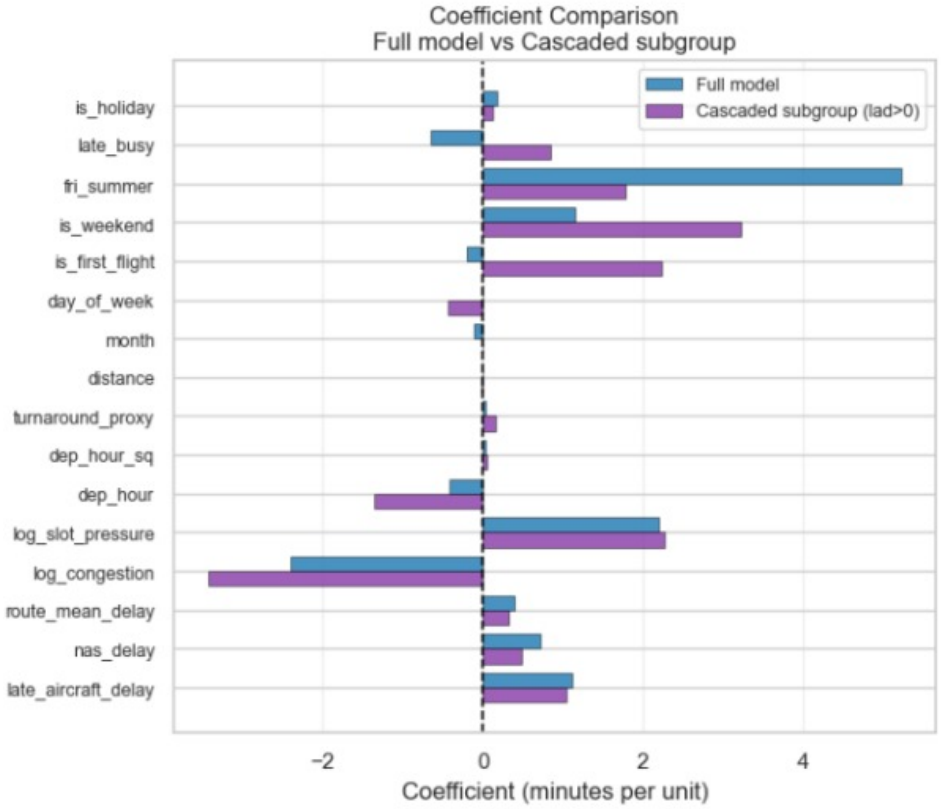


Figure 7: Comparision of OLS Coefficients for Models trained on the Full Dataset vs Cascaded Subgroup

Using this regression as a base, we used K-Means Clustering ($k = 3$) on the carrier fixed effects to sort airlines by their structural delay DNA. We used strong fixed effects from our cleaned regression to find three different operational profiles. This gave us a strategic Intervention Matrix for improving performance. This stage changed our view of carriers from a single "average" view to a more detailed understanding of Root Cause Profiles, which separates "Carrier-Owned" faults from "External Cascade" vulnerabilities.

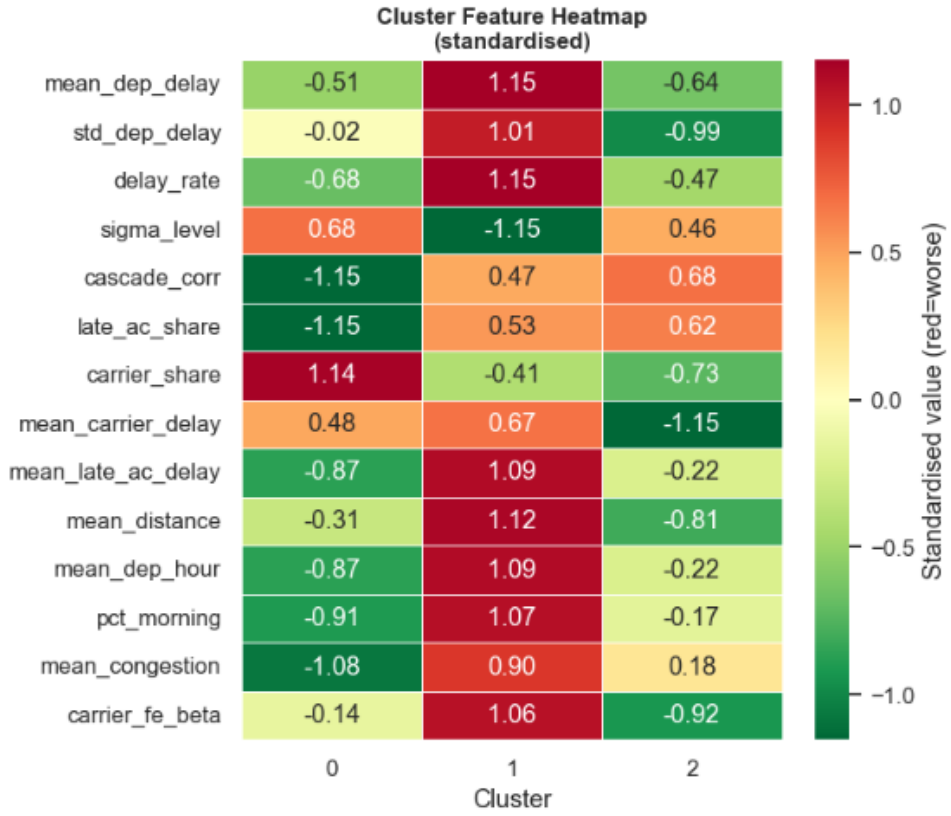


Figure 8: Cluster Analysis - Carrier Fixed Effects

The diagnostic results show that different carriers need different fixes. For example, the Cluster Feature Heatmap shows that "Structurally Poor" carriers (Cluster 1) are red-coded for almost all negative metrics, such as high delay rates and sigma levels.

On the other hand, our "Cascade-Prone" group (Cluster 2) has lower internal fault shares but is very sensitive to delays, as shown by the big jump in Late-AC Share on the Cluster Radar Profile (Fig. 9). At the same time, "High Performers" (Cluster 0) have negative fixed effects and high sigma levels, which puts them in the "best-in-class" top-left quadrant of our diagnostics. We showed that airline performance follows measurable, clusterable patterns by putting these carriers into groups based on statistics. This makes it possible to make targeted changes, like building operational buffers for carriers that are sensitive to cascades or doing deep structural overhauls for carriers with high fixed-effect coefficients.

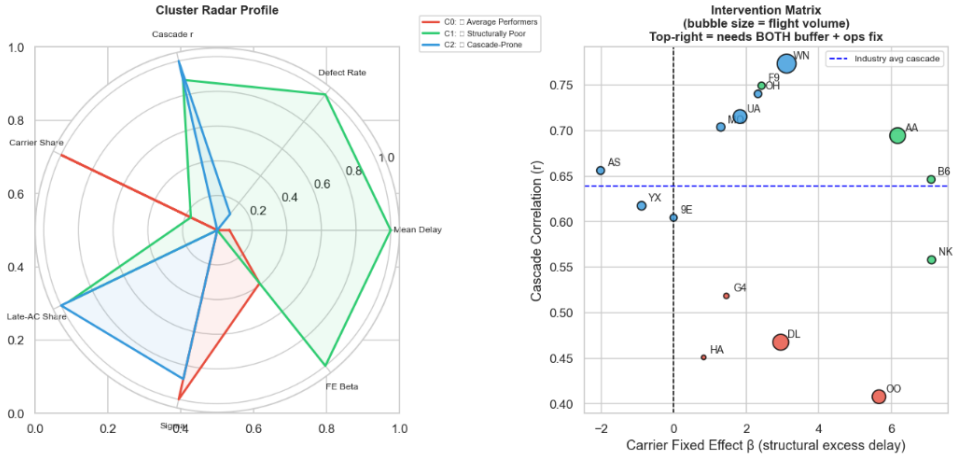


Figure 9: Strategic Intervention Matrix

5 Improve Phase

Approach: Since 74.9% of delay minutes are controllable and the cascade analysis shows that early-rotation delays propagate through the day's schedule, the primary operational lever is schedule buffer time, adding slack between rotations to absorb upstream delays before they cascade.

5.1 Linear Programming Formulation

Decision variables represent buffer minutes allocated per flight for each of the top 6 carriers (WN, AA, DL, UA, OO, NK). The objective maximises total weighted delay reduction, where weights reflect each carrier's share of total flight volume. The total buffer budget is constrained to 5 minutes per flight, with a fairness floor of 0.3 minutes minimum per airline.

Optimal Allocation Result:

Table 3: Optimal Buffer Allocation

Airline	Weight	Buffer Allocated	Delay Saved
WN	27.50%	3.5 min	3.5 min
AA	18.90%	0.3 min	0.3 min
DL	19.60%	0.3 min	0.3 min
UA	–	0.3 min	0.3 min
OO	–	0.3 min	0.3 min
NK	–	0.3 min	0.3 min

Southwest receives the bulk of the buffer (3.5 minutes) given its 27.5% share of total flights, saving 1 minute at WN yields more industry-wide benefit than at any other carrier. Total estimated savings across the top 6 airlines: 6.03 million delay-minutes with a 5 min/flight budget. But note that the severity-weighted formulation (prioritising American, which has the highest mean delay at 20.58 min) would shift allocation toward AA, the appropriate weighting depends on whether the objective is aggregate volume reduction or worst-carrier correction.

6 Control Phase

X-Bar & R Charts: SPC charts were constructed using batches of 30 consecutive flights per airline to monitor process stability. American Airlines' center line of 20.58 minutes already exceeds the CTQ threshold of 15 minutes, meaning the average batch of 30 consecutive AA flights is itself a defect. Both carriers show wide control bands (35+ minutes), confirming highly variable, unstable processes with significant special-cause variation rather than mere random noise.

Table 4: X-Bar Control Chart Results for the two highest-volume carriers:

Airline	Center Line ($\bar{X}$)	UCL	LCL	% Out-of-Control
WN	11.71 min	29.60 min	-6.17 min	6.40%
AA	20.58 min	–	–	Avg batch already exceeds CTQ

P-Chart - Monthly Defect Rate: The p-chart monitors the proportion of delayed flights month by month against statistically computed control limits ($UCL \approx 20.10\%$, $LCL \approx 19.78\%$).

- Overall $\bar{p}$: 19.94%
- Above UCL (out of control, too many delays): January, March, May, June, July, August, December
- Below LCL (better than baseline): February, April, September, October, November
- Best month: October at 13.0%
- Worst month: July at 28.7%
- Seasonal swing: 15.6 percentage points

The p-chart makes clear that summer and holiday months (July, December) represent genuine special-cause deterioration, not just random variation – they require targeted interventions such as additional reserve crew deployment and proactive schedule padding rather than general process improvement.

7 Conclusions

The U.S. commercial aviation system operated at a sigma level of 2.34σ in 2024, every single carrier below 3σ, with C_{pk} below 0.1 across the board. The analysis identifies three tiers of actionable insight:

1. **Controllable Delays:** 74.9% of delay minutes are controllable, concentrated in late aircraft cascades (40.4%) and carrier operational failures (34.5%).
2. **Cascade Mechanism:** The cascade mechanism is statistically proven ($r = 0.639$) and nearly deterministic at the departure-to-arrival level ($r = 0.971$), making early-rotation buffer time the highest-leverage single intervention.



Figure 10: X-Bar and R Control Charts

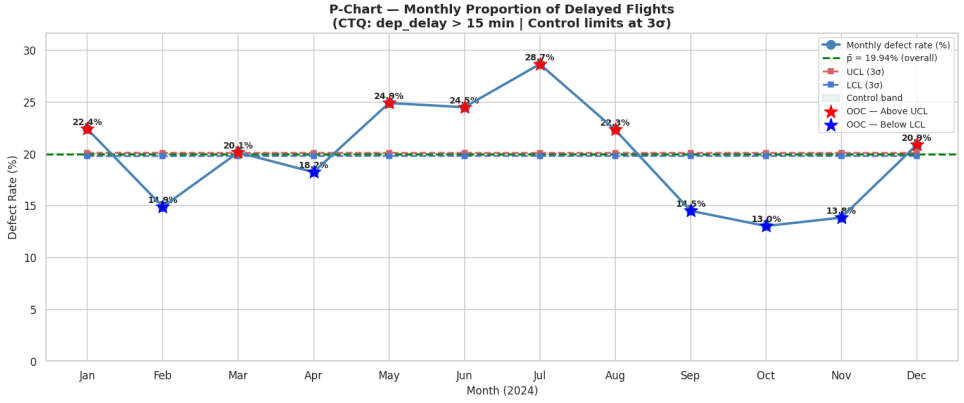


Figure 11: P-Chart

3. **Pareto Concentration:** Pareto concentration is strong - fixing six carriers (WN, AA, DL, UA, OO, NK) addresses 75% of all delayed flights industry-wide.

The LP model estimates that a 5-minute-per-flight buffer budget across the top 6 carriers would save 6.03 million delay-minutes. To reach 3σ, the industry requires approximately a 50% reduction in its current delay rate - an ambitious but structurally grounded target, achievable through phased improvements in turnaround discipline, reserve crew deployment during peak months, and ATC coordination at the highest-cascade airports.

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